1. Introduction

1.1 Project Abstract

The Show Booking Web Application is a comprehensive and user-friendly platform that allows users to browse, book, and manage tickets for various Shows, including movies, concerts, and theater productions. The application is designed to streamline the booking process, enhance customer experience, and provide a secure and efficient way to manage Show-related transactions. By leveraging modern web technologies, the platform ensures seamless navigation, secure transactions, and real-time updates for users, Show organizers, and administrators.

1.2 Objectives

The primary objective of this project is to develop a Show booking system that simplifies ticket reservations while ensuring data integrity and security. The platform aims to provide an intuitive interface for users to search for Shows, book tickets, and manage their reservations efficiently. Additionally, Show organizers can create and update Show listings, track sales, and analyze Show performance. Administrators have access to advanced reporting and monitoring tools to oversee the platform's operations.

2. System Modules & Implementation

2.1 User Module

The User Module is designed to provide a smooth and interactive experience for customers booking tickets. Users can register on the platform by providing their details, and once logged in, they can explore available Shows. The system offers various filtering options based on date, genre, and location to help users find their preferred Shows quickly. After selecting a Show, users can choose their seats, proceed with payment, and receive a confirmation. Additionally, users can manage their bookings by viewing past reservations, downloading e-tickets, and requesting cancellations or refunds if needed.

2.2 Show Organizer Module

Show organizers play a crucial role in the system as they are responsible for managing Show listings and tracking ticket sales. After logging into their accounts, organizers can create new

Show listings by specifying details such as the Show title, date, time, venue, ticket price, and seating availability. The system allows real-time updates to Show information, ensuring users always have access to the latest details. Organizers can also monitor ticket sales through a dedicated dashboard, providing insights into Show performance and audience engagement.

2.3 Admin Module

The Admin Module is designed to maintain the overall integrity and smooth operation of the platform. Administrators have access to a dashboard where they can monitor all user and organizer activities. They are responsible for managing user accounts, addressing queries, and ensuring compliance with platform policies. Additionally, the system generates detailed reports on ticket sales, user demographics, and Show popularity, helping administrators make informed decisions to improve the platform. Security measures such as fraud detection, encryption, and access control are also managed by the admin panel.

3. Project Flow

3.1 User Registration & Login

Users begin by creating an account on the platform, providing necessary details such as name, email, and password. The system performs validation checks before storing user information securely in the MySQL database. Users can log in using their credentials, and authentication is managed using Spring Security to ensure data protection.

3.2 Show Browsing & Selection

Once logged in, users can browse Shows through an interactive Angular-based interface. The platform allows users to filter Shows based on various parameters such as genre, date, and location. Each Show listing provides detailed information, including time, venue, available seats, and ticket prices, helping users make informed booking decisions.

3.3 Ticket Booking & Payment

Users can select their preferred seats using an interactive seating chart displayed on the Show details page. After finalizing seat selection, they proceed to the payment gateway, where they can securely enter payment details. The backend, developed using Spring Boot, processes transactions securely, ensuring successful booking confirmation. Once the payment is completed, users receive an electronic ticket and a confirmation email.

3.4 Booking Management

The platform allows users to access their booking history, download tickets, and manage bookings efficiently. Users can request cancellations or refunds based on platform policies. The system also sends automated notifications to users regarding booking status updates, reminders for upcoming Shows, and changes in Show details.

3.5 Show Organizer Operations

Show organizers use the platform to create and manage Shows efficiently. They can add new Show listings by providing all necessary details, ensuring users have up-to-date information. Organizers have the flexibility to update event details, cancel Shows if needed, and track real-time ticket sales through their dashboard. These features help organizers plan and optimize their events effectively.

3.6 Administrative Functions

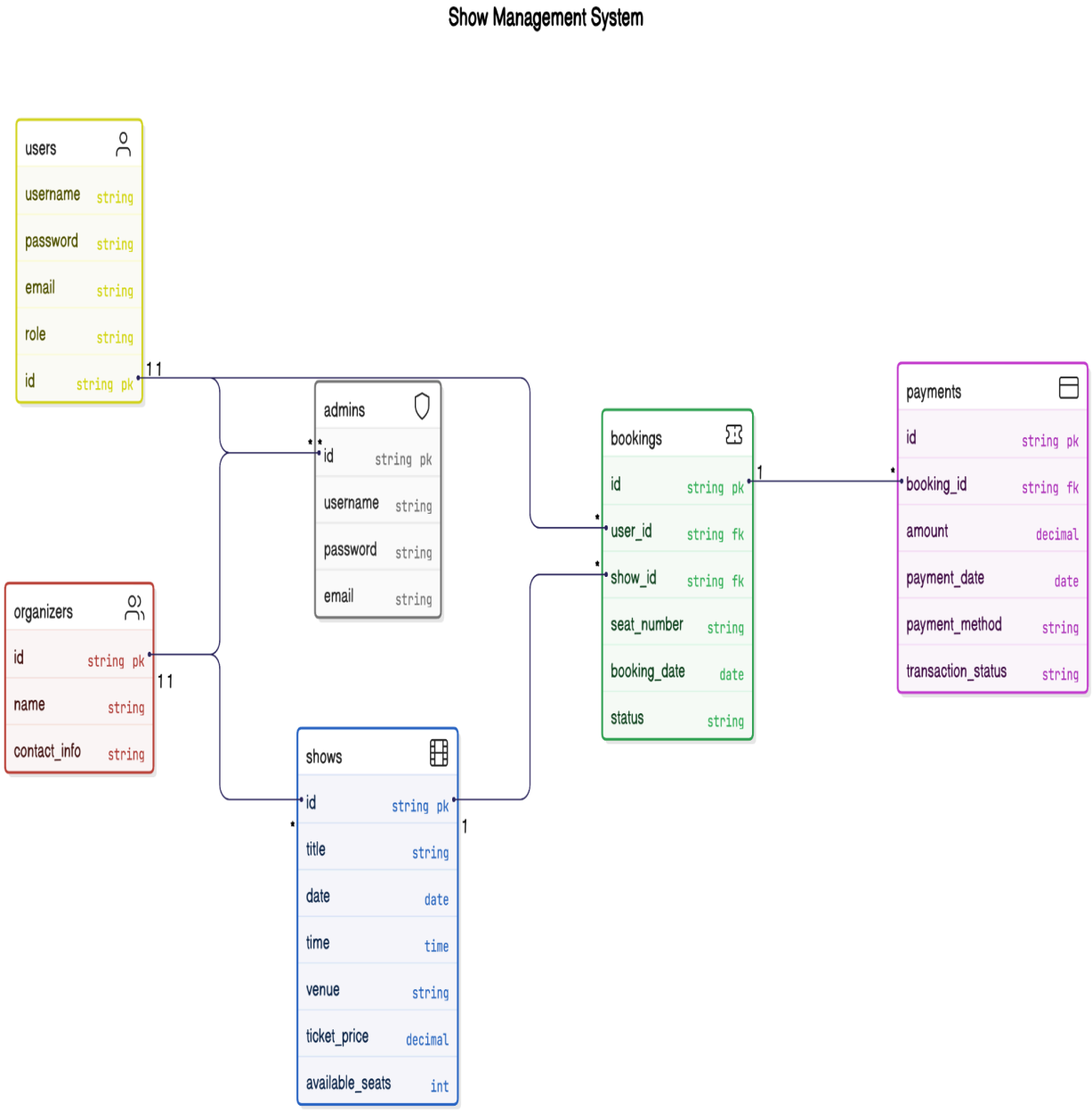
The administrator panel provides comprehensive control over platform operations. Admins oversee user and organizer activities, manage disputes or issues, and ensure that Shows adhere to platform guidelines. They can generate reports on various performance metrics, such as total sales, most popular Shows, and user engagement trends. The system also integrates security measures to protect sensitive data and prevent fraudulent transactions.

4. Technology Stack

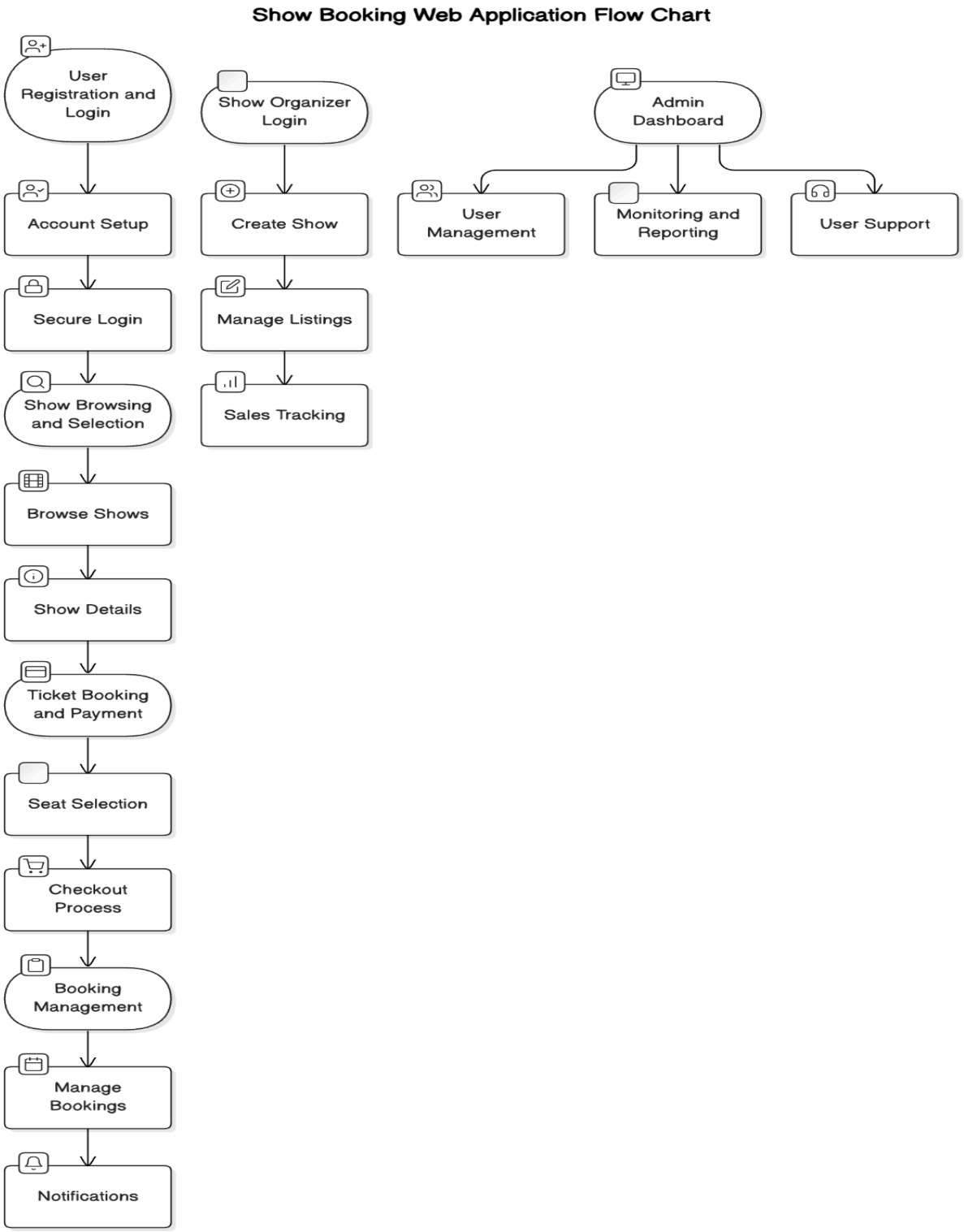
The application is built using a combination of robust frontend and backend technologies to ensure performance, scalability, and security.

- **Frontend Technologies**: HTML, CSS, JavaScript, and Bootstrap are used to create a responsive and visually appealing user interface.
- **Frontend Framework**: Angular is used to enhance interactivity and provide a smooth user experience.
- **Backend Technologies**: Java-based Spring Boot framework handles backend logic, while Spring MVC and Hibernate facilitate data management.
- **Database**: MySQL is used to store user, Show, and booking data efficiently.
- **Security Measures**: The system uses HTTPS, encrypted data storage, and Spring Security for authentication and access control.

5. ER Diagram


Show Management System
users
username string
password string
email string
role string
id string pk
admins
id string pk
username string
password string
email string
organizers
id string pk
name string
contact_info string
shows
id string pk
title string
date date
time time
venue string
ticket_price decimal
available_seats int
bookings
id string pk
user_id string fk
show_id string fk
seat_number string
booking_date date
status string
payments
id string pk
booking_id string fk
amount decimal
payment_date date
payment_method string
transaction_status string
1
1
1
1
1
1

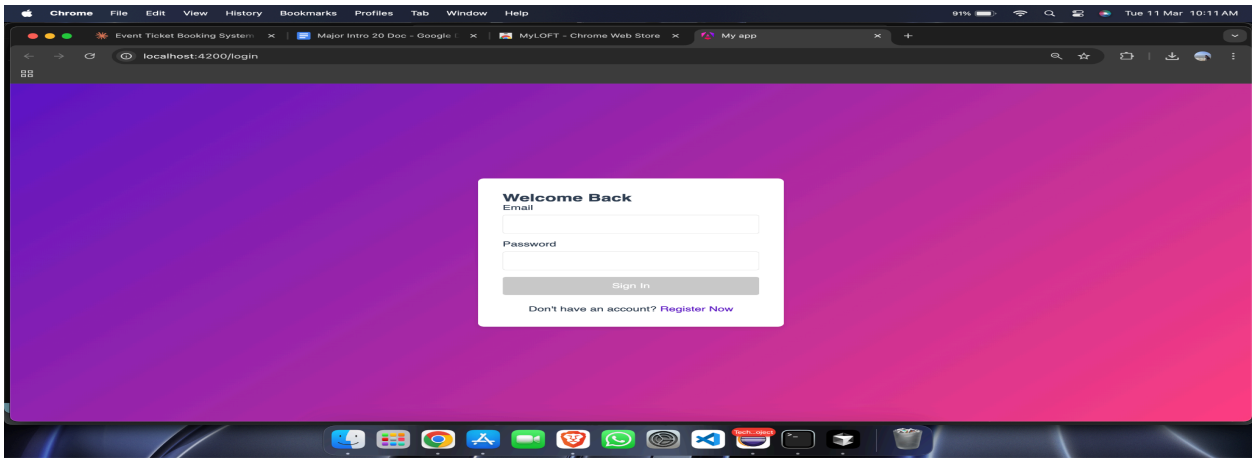
6. Use Case Diagram

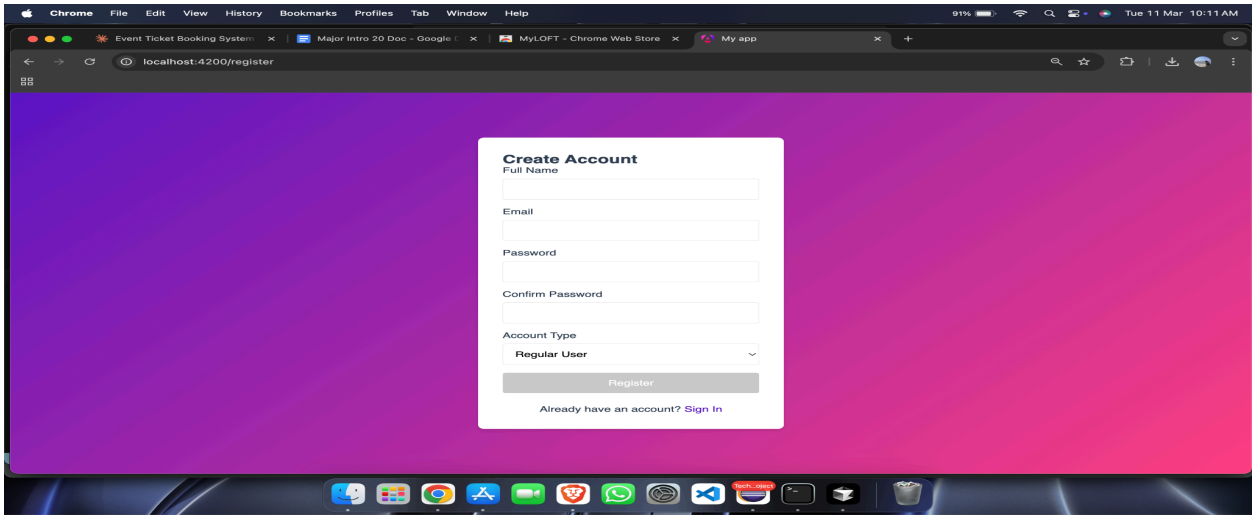


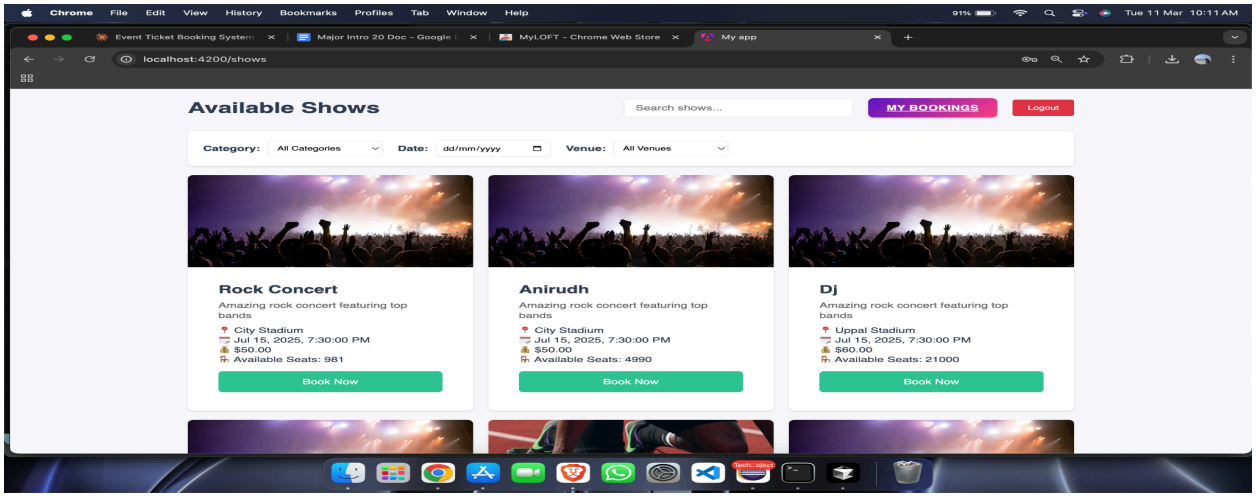
7. Flowchart

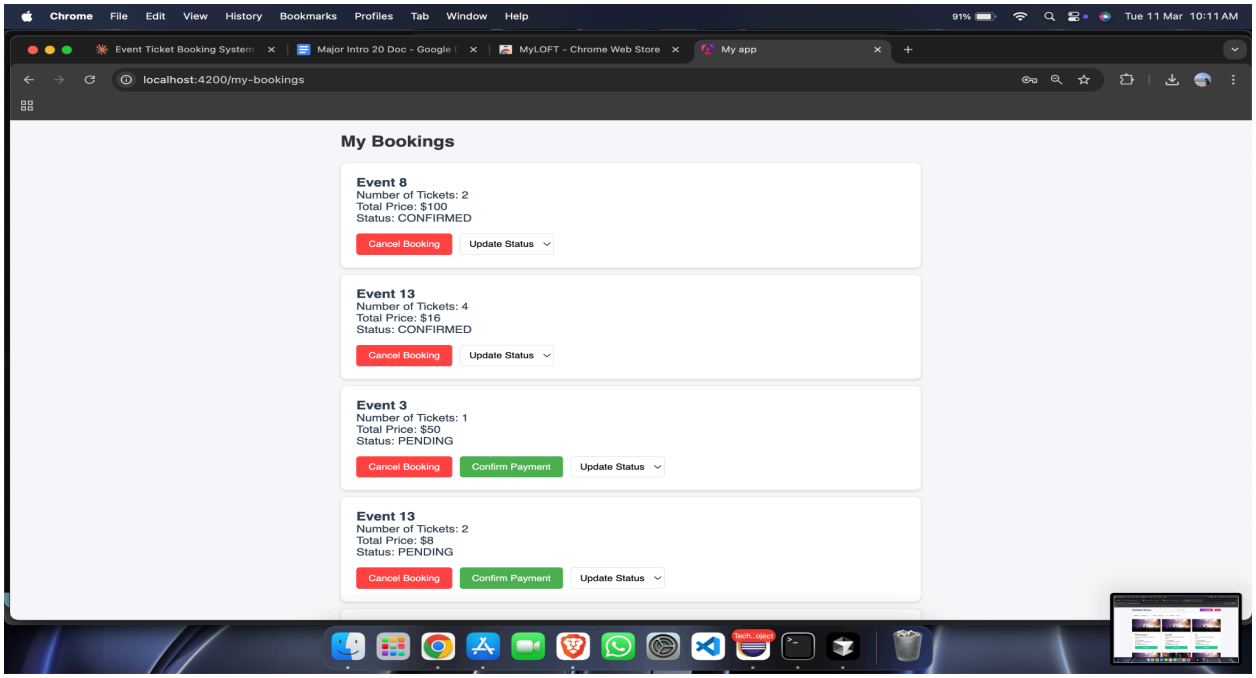

Start
Begin the process with initial preparations and gather necessary information.
Account Setup
Users provide necessary details to create an account. Validation checks are performed before storing data in MySQL.
Secure Login
Users log in using credentials verified via Spring Security.
Browse Shows
Users utilize the Angular-powered frontend to filter and select Shows based on preferences.
Show Details
Users can view detailed information about Shows, including date, time, venue, and available seats.
Seat Selection
Users select their preferred seats from an interactive seating chart.
Checkout Process
Users proceed to payment, entering payment details secured by Spring Boot.
Manage Bookings
Users can access their booking history, download tickets, or request cancellations and refunds.
End
Conclude the process with all bookings managed.

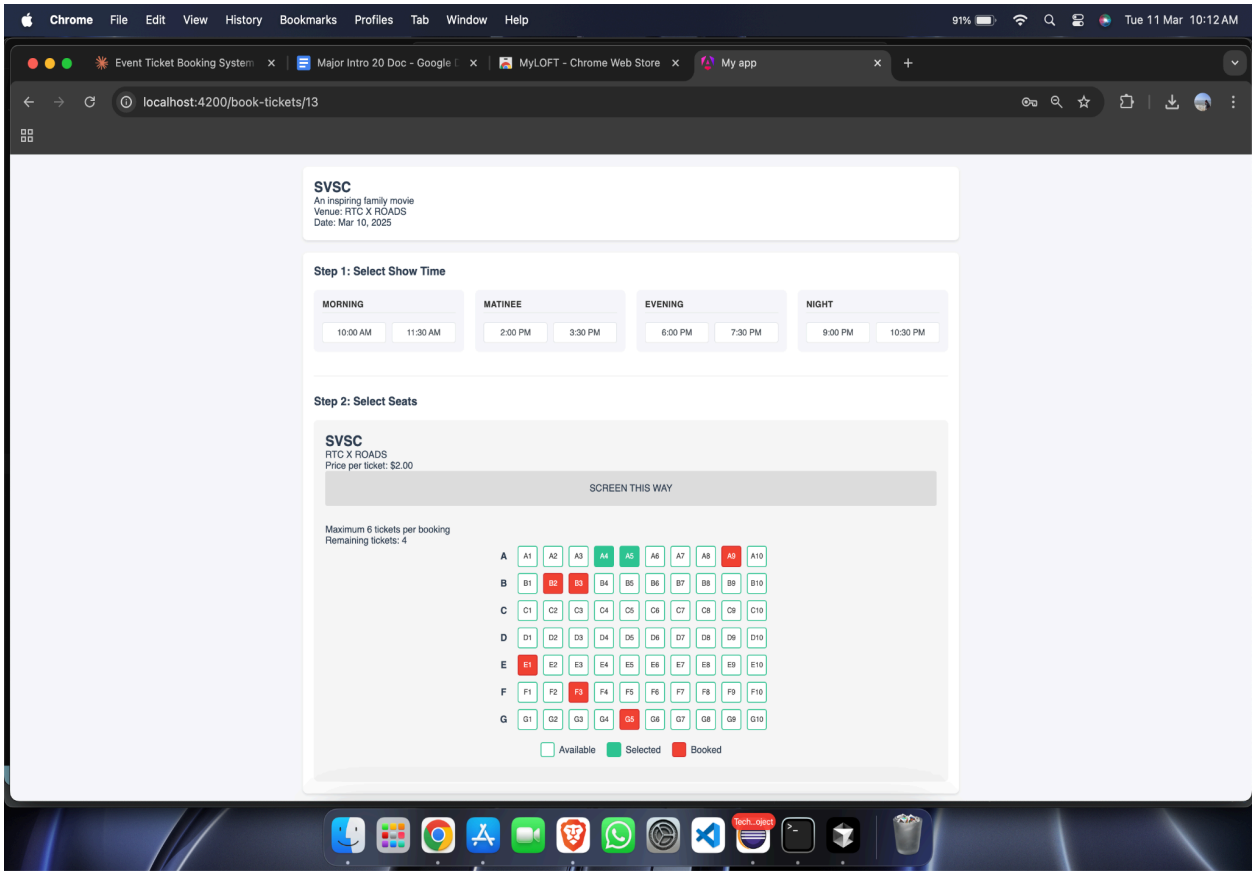
8.Output Screenshots

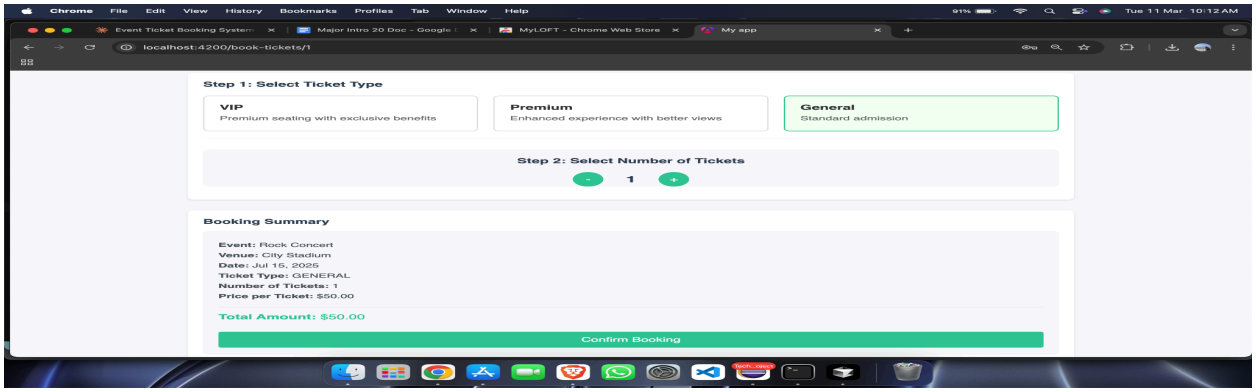


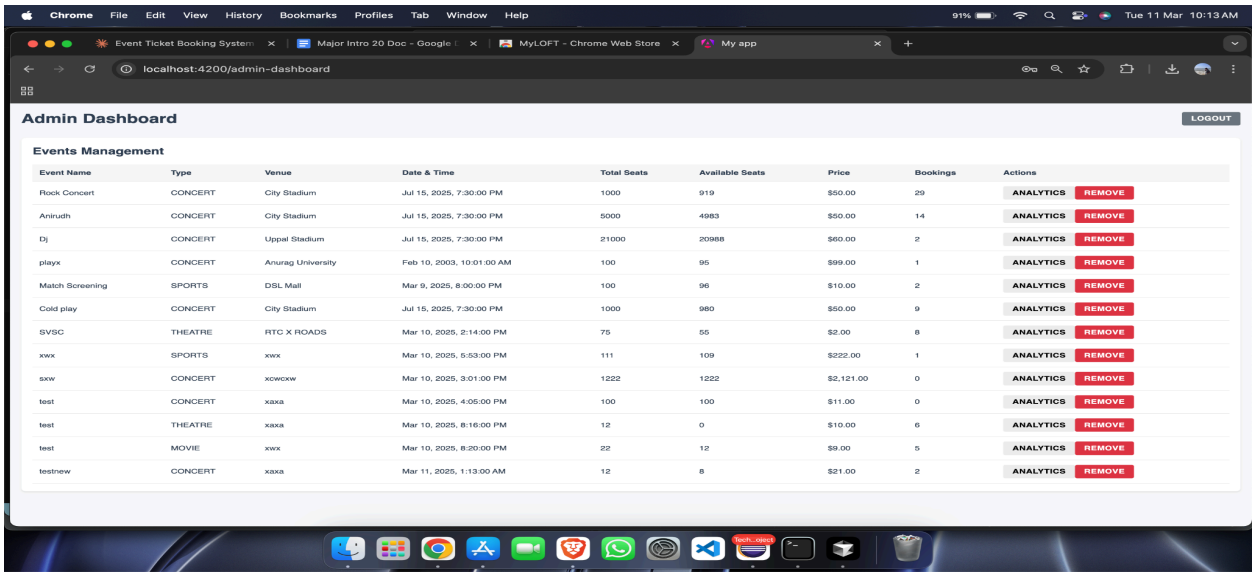





Chrome File Edit View History Bookmarks Profiles Tab Window Help
91% Tue 11 Mar 10:11 AM
Event Ticket Booking System
Major Intro 20 Doc - Google
MyLOFT - Chrome Web Store
My app
localhost:4200/my-bookings
My Bookings
Event 8
Number of Tickets: 2
Total Price: $100
Status: CONFIRMED
Cancel Booking
Update Status
Event 13
Number of Tickets: 4
Total Price: $16
Status: CONFIRMED
Cancel Booking
Update Status
Event 3
Number of Tickets: 1
Total Price: $50
Status: PENDING
Cancel Booking
Confirm Payment
Update Status
Event 13
Number of Tickets: 2
Total Price: $8
Status: PENDING
Cancel Booking
Confirm Payment
Update Status


Chrome File Edit View History Bookmarks Profiles Tab Window Help
91% Tue 11 Mar 10:12 AM
Event Ticket Booking System
Major Intro 20 Doc - Google
MyLOFT - Chrome Web Store
My app
localhost:4200/book-tickets/13
SVSC
An inspiring family movie
Venue: RTC X ROADS
Date: Mar 10, 2025
Step 1: Select Show Time
MORNING
10:00 AM
11:30 AM
MATINEE
2:00 PM
3:30 PM
EVENING
6:00 PM
7:30 PM
NIGHT
9:00 PM
10:30 PM
Step 2: Select Seats
SVSC
RTC X ROADS
Price per ticket: $2.00
SCREEN THIS WAY
Maximum 6 tickets per booking
Remaining tickets: 4
A A1 A2 A3 A4 A5 A6 A7 A8 A9 A10
B B1 B2 B3 B4 B5 B6 B7 B8 B9 B10
C C1 C2 C3 C4 C5 C6 C7 C8 C9 C10
D D1 D2 D3 D4 D5 D6 D7 D8 D9 D10
E E1 E2 E3 E4 E5 E6 E7 E8 E9 E10
F F1 F2 F3 F4 F5 F6 F7 F8 F9 F10
G G1 G2 G3 G4 G5 G6 G7 G8 G9 G10
Available
Selected
Booked





9. Conclusion

The Show Booking Web Application offers a seamless and secure platform for users to book and manage entertainment event tickets. With its intuitive design, real-time updates, and efficient Show management features, the platform enhances the overall booking experience. By leveraging Angular for frontend development and Spring Boot for backend services, the application ensures high performance and data security. The system's scalability and advanced reporting features also benefit Show organizers and administrators, making it a robust solution for modern entertainment booking needs.